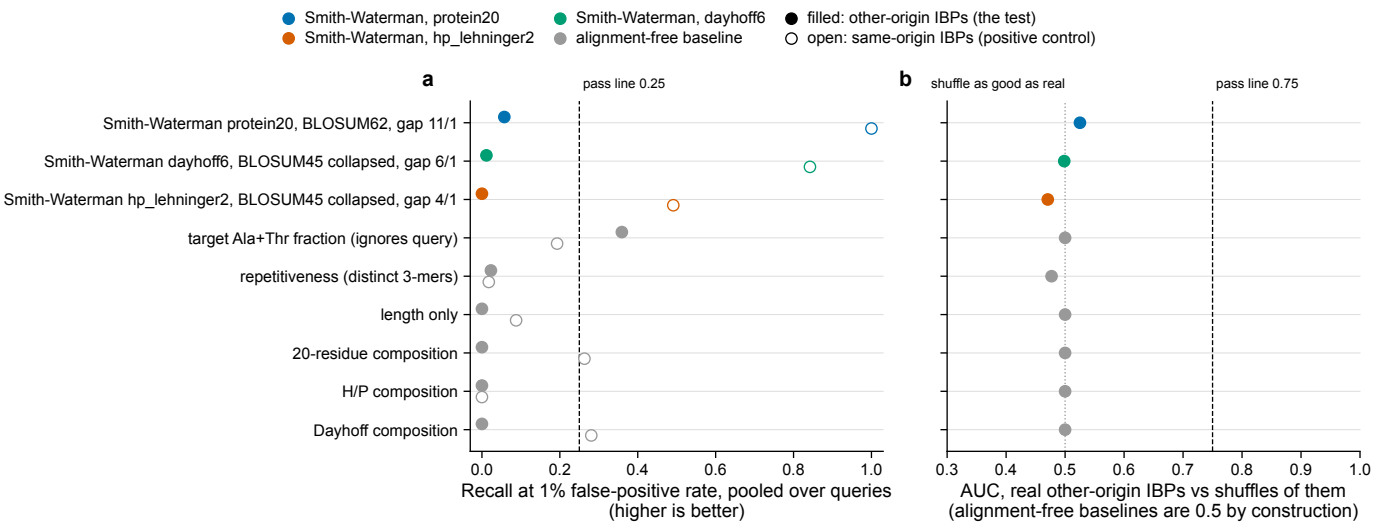
Smith-Waterman, protein20
Smith-Waterman, hp_lehninger2
Smith-Waterman, dayhoff6
alignment-free baseline
filled: other-origin IBPs (the test)
open: same-origin IBPs (positive control)
a
pass line 0.25
Smith-Waterman protein20, BLOSUM62, gap 11/1
Smith-Waterman dayhoff6, BLOSUM45 collapsed, gap 6/1
Smith-Waterman hp_lehninger2, BLOSUM45 collapsed, gap 4/1
target Ala+Thr fraction (ignores query)
repetitiveness (distinct 3-mers)
length only
20-residue composition
H/P composition
Dayhoff composition
0.0
0.2
0.4
0.6
0.8
1.0
Recall at 1% false-positive rate, pooled over queries
(higher is better)
b
shuffle as good as real
pass line 0.75
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
AUC, real other-origin IBPs vs shuffles of them
(alignment-free baselines are 0.5 by construction)